

Intro to Transfer Functions

EGR 345

GVSU

Outline

- review
- intro to transfer functions
 - what is a transfer function?
 - how do we find transfer functions?
 - why are they so powerful?
 - what is an impulse response?
 - what is a step response?

Review

- what is a dynamic system?

Review

- what is a dynamic system?
- why is Dr. Krauss obsessed with mass/spring/damper systems?

Review

- what is a dynamic system?
- why is Dr. Krauss obsessed with mass/spring/damper systems?
- where can we find MSD systems in real-life?

What is a transfer function?

How do we find transfer functions?

What can we do with transfer functions?

What is an impulse response?

- and how do we find it?
 - “by hand”
 - with python

What is a step response?

- and how do we find it?
 - “by hand”
 - with python

What is a step function?

- how is $1(t)$ different from just the number 1?